

Square Roots & Cube Roots

Year 7 Maths · Unit 1.6 · by Mr Bowen

KEY WORDS

Square number

số chính phương

a number multiplied by itself

$$5^2 = 5 \times 5 = 25$$

Square root $\sqrt{\quad}$

căn bậc hai

the number that was multiplied by itself

$$\sqrt{25} = 5, \text{ because } 5 \times 5 = 25$$

Cube number

số lập phương

a number multiplied by itself twice

$$2^3 = 2 \times 2 \times 2 = 8$$

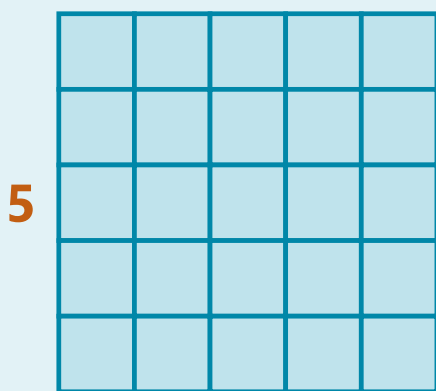
Cube root $\sqrt[3]{\quad}$

căn bậc ba

the number that was multiplied by itself twice

$$\sqrt[3]{8} = 2, \text{ because } 2 \times 2 \times 2 = 8$$

SEE IT

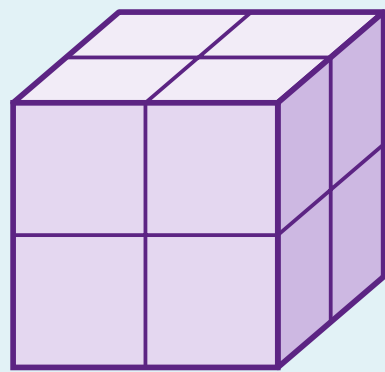


5

5

$$5 \times 5 = 25$$

$$\sqrt{25} = 5$$



2

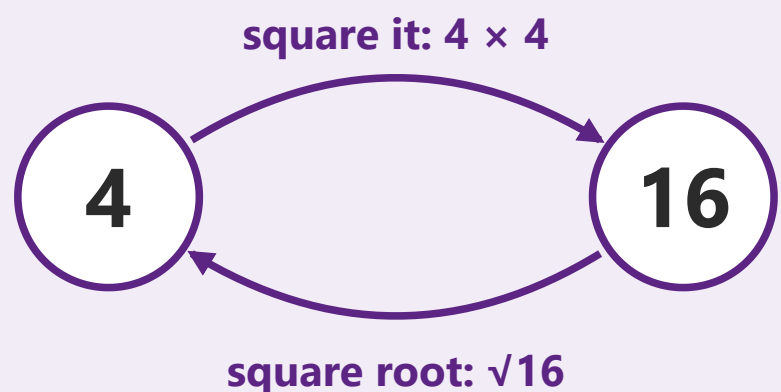
$$2 \times 2 \times 2 = 8$$

$$\sqrt[3]{8} = 2$$

The names come from the pictures.

HOW IT WORKS

Squaring and square rooting
undo each other.



Ask yourself:

What number $\times$ itself makes this?

EXAMPLES

1. Find $\sqrt{49}$

Which number $\times$ itself = 49?

$$7 \times 7 = 49, \text{ so } \sqrt{49} = 7$$

2. Mr Bowen's patio

225 square stones make one big square.

How many stones along one side?

$$\sqrt{225} = 15, \text{ because } 15 \times 15 = 225$$

15 stones along each side

CAREFUL!

$$\cancel{\sqrt{16} = 8} \quad \times$$

$$\sqrt{16} = 4, \text{ because } 4 \times 4 = 16 \quad \checkmark$$

Square root is not $\div 2$.

TRY IT!

What is $\sqrt{100}$?

What is $\sqrt[3]{27}$?

Lift me!

answers under here

Learn these: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144